

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Computer Science and Engineering) SEMESTER - 7 Summer 2025 ( Supply. )**

**Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 7**

**Subject Code & Name: BTCOE703C - BIG DATA ANALYTICS**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

1. What is the main challenge in handling Big Data?  
A. High storage cost      B. Data consistency  
C. Scalability and processing speed      D. User interface design  
Which of the following is a Big Data distribution package?  
A. MATLAB      B. Hadoop  
C. Oracle      D. Eclipse  
In Spark, the main abstraction for data is called:  
A. Hive      B. RDD  
C. Node      D. Stream
4. The CAP Theorem does NOT include:  
A. Consistency      B. Accuracy  
C. Availability      D. Partition tolerance
5. Which of the following tools is widely used for Big Data streaming?  
A. Hive      B. Kafka  
C. Pig      D. Storm  
What is a key requirement for Big Data streaming systems?  
A. Batch processing      B. Offline analysis  
C. Low latency and high throughput      D. SQL support  
Mahout is a library for:  
A. Data storage  
B. Stream processing  
C. Machine learning  
D. Visualization
8. Which algorithm is used for classification in Mahout?  
A. K-means      B. Apriori      C. Naive Bayes      D. PageRank

- 9 GraphX is a component of:  
 A. Cassandra  
 B. Spark  
 C. Mahout  
 D. HDFS
- 10 MongoDB is a:  
 A. Relational DBMS      B. Document-oriented NoSQL database  
 C. Column-store database      D. Key-value store

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- 11 In MongoDB, data is stored in:  
 A. Tables      B. Collections  
 C. Files      D. Buckets
- 12 Which command is used to insert a document in MongoDB shell?  
 A. add()      B. put()  
 C. insert()      D. update()

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- Q2. Solve the following.
- A) Describe the Big Data Stack and its components. 6
- B) Explain enabling technologies that support Big Data. 6
- Q3. Solve the following.
- A) Explain the CAP Theorem and trade-offs in distributed systems. 6
- B) Discuss the differences between ACID and BASE models. 6
- Q4. Solve Any Two of the following.
- A) What are the challenges in real-time Big Data processing? 6
- B) Compare batch processing vs. stream processing. 6
- C) Describe Kafka and its role in real-time pipelines. 6
- Q5. Solve Any Two of the following.
- A) Compare Mahout and Spark MLlib for Big Data ML. 6
- B) Explain k-means clustering algorithm in Big Data context. 6
- C) What is GraphX? How does it support graph processing? 6
- Q6. Solve Any Two of the following.
- A) Explain collections and documents in MongoDB with examples. 6
- B) Describe the principles of schema design in MongoDB. 6
- C) Explain the key features of MongoDB. 6

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